Nicole Lee

nicole.lee@berkeley.edu | linkedin.com/in/nicolehylee/ | github.com/nicoleleehy1 | nicole-hy-lee.vercel.app/

Education

University of California, Berkeley Expected Graduation: May 2028
B.S. in Electrical Engineering and Computer Science, B.S. in Bioengineering

- **Hong Kong Scholarship for Excellence Scheme Scholar** - USD$155,000 scholarship to study at UC Berkeley
- **Coursework:** Data Structures & Algorithms, Computer Networks, Web Development, Computer Architecture, Machine Learning, Databases, Intractable Algorithms, Internet Architecture, User Design, Computational Biology
- **Activities:** Cal Hacks (Director), Web Development at Berkeley (Teaching Assistant), Notion Campus Leader, Society for Women Engineers

Skills

Languages: Typescript, C/C++, Java, Python, Swift, Objective-C, Go, Rust, SQL, RISC-V
Frontend & Systems: React, Next.js, Linux, Git, Bash, CI/CD, Distributed Systems
Backend: FastAPI/REST API, Node.js, Express.js, PostgreSQL, Redis, MongoDB, AWS (EC2, S3, Lambda), Azure, Docker, Kafka
AI/ML: TensorFlow, PyTorch, LangGraph, MediaPipe, Hugging Face, YOLOv12

Experience

Apple | Software Engineer Intern May 2026 - Aug 2026

- **Built a full MCP Evaluation platform** across **450+ servers**, saving **$700,000+/year** by generating **200+ schema-aware** ground truth queries per run and automating live tool calls to cut **per-eval time 18x** (6+hrs → 20mins)
- **Architected fault-tolerant tracing proxy** intercepting live MCP traffic with response normalization and persistent PostgreSQL state, **collapsed 100+ correlated failures into <10 root causes**, cutting **triage 40%**.
- **Engineered hybrid scoring** with deterministic tool-call/JSON diffing + LLM-as-judges; extracted Langfuse traces + A/B runs to analyze routing failures and recommend tool description/routing changes optimizing accuracy, latency & cost

Wedge (YC S25) | Forward Deployed Engineer Intern Aug 2025 – Dec 2025

- Built a clinician analytics platform for LA General Medical Center (600-bed Level I trauma center) using React, TypeScript, and GraphQL, surfacing EHR and scheduling across **30K+ annual discharges** to improve bed turnover.
- Architected **20+ reusable frontend components** and **REST data pipelines** for FHIR/HL7 EHR data, adding client-side caching, conflict detection, and utilization scoring to support low-latency analytics under concurrent workloads.

Anchor Logics | Software Engineer Intern Jun 2025 – Aug 2025

- **Led 5 engineers** to ship a real-time telemetry platform that analyzes **3D IMU data streamed from IoT vests** used by **50+ ALS/Parkinson's patients**, surfacing data visualizations, timeline scrubbers, and playback controls.
- Built CV and data infrastructure with YOLOv12, OpenPose, DeepSORT, AWS S3, Redis, and PostgreSQL, improving **keypoint consistency 25%**, cutting occlusion loss 35%, and supporting **2,000+ models with sub-100ms retrieval**.

Cal Hacks | Director Aug 2024 – Present

- **Built and deployed 3 production websites** including public site, hive system, and live hacker portal (React, Typescript, Supabase, PostgreSQL) and **published iOS app** (C, Swift) for the largest collegiate hackathon, serving **4,000+ hackers and 60+ sponsors**.
- Built an **automated sponsorship pipeline** reaching **2,000+ companies** and **14,000+ contacts**, raising **$1,250,000+**.

DigPath | Software Engineer Intern Apr 2025 – Jun 2025

- **Deployed a SageMaker + Meta SAM cell segmentation pipeline** with GPU-accelerated inference across 500+ whole-slide pathology images (70GB dataset); **reduced manual annotation** turnaround from days to under **2 hours** per case.
- **Optimized serverless platform** (AWS Amplify, Lambda, DynamoDB) tracking pathology image ingestion, storage and case volume.

University of California, San Francisco | Undergraduate ML Researcher Aug 2024 – Feb 2025

- Built a **Python RAG pipeline** indexing **2,000+ neuroscience papers** using sentence-transformer embeddings and FAISS
- Implemented document chunking, metadata-aware retrieval, and Azure OpenAI answer generation with a Gradio application to source citations.

Projects

Aster [Link] [GitHub & Research] | *JPython, TypeScript, FastAPI, PyTorch, RDKit, LangGraph*

- Integrated Boltz-2 protein–ligand modeling and a five-model D-MPNN ensemble into a molecular prioritization workflow; built a separate evaluation harness comparing seven selection policies across 252 retrospective replays and six targets.
- Implemented asynchronous scoring and transactional campaign updates with revision checks, idempotent batch commits, and immutable prediction snapshots to reject stale recommendations and preserve experiment history.

CodeMRI [GitHub] | *Python · TypeScript · Tree-sitter · FastAPI*

- Built a **static-analysis + context compiler for coding agents** parsing repository-wide codebase ASTs into persistent architecture and symbol graphs surfacing call/import dependencies, source navigation, and change-impact analysis
- Compiled change-scoped context **reducing repository-reading tokens by 83%** and enabling graph-guided code modifications.

Flusk (Flush + Disk LSM Tree) [Link] [GitHub] | *Java 17 · Maven · Guava · Snappy · JUnit 5 · Railway*

- Implemented an **LSM-tree storage engine** with write-ahead logging, MemTables, SSTables, and four-tier compaction with tombstone garbage collection; added Bloom filters and sparse indexes for key lookup.